THE PROBLEM

Robots now act on what an AI tells them. Nobody can see what it was shown, who paid, or whether the decision was real.

Camera images go to a model you can’t inspect. When the cloud AI is slow or down, a robot either guesses or freezes. And the record of what it decided lives in the operator’s logs.

WHAT WE BUILT

Semmal: a robot that can't pass a checkpoint without 0G

Drive
ROS 2 on the robot

→

Hire + pay
an AI agent, in native 0G

→

Decide on 0G
private TEE model, request ID

→

Check + act
bounded actions only

→

Record
signed episode on 0G Storage

0G is the brain at each checkpoint, the money and the memory. Motion and safety stay on the robot.

DELIVERABLE 1 · PRIVATE 0G DECISION IN THE LOOP, WITH ITS REQUEST ID

The customer sees every call, in full

```
0gm-1.0-35b-a3b · private (TEE providers only)
0G request ID   985498df-94a2-4e3d-9f1d-21ffe9f8a472
Time            24,464 ms
Billed to       the agent's own Router key · 0.5 USD/day
cap
Answer          "toys on floor" · confidence 0.95
                → clear_and_continue
Checked         ✓ exact format  ✓ allowed action  ✓ ≥
0.6
Did             picked the toys up, put them in the bin
```

No silent downgrade

A model without TEE is never used for a private job.

Advisory only

The model picks one of five actions; it never touches motors, money or keys.

DELIVERABLE 2 · THE ROBOT PAYS ITS OWN AGENTS IN 0G

A robot with its own wallet and budget

Customer → robot
pays the robot's signed quote, after it arrives

→

Robot → agent owner
one transfer per job, checked on chain before 0G is called, never twice

→

Agent → 0G PC
its own capped key

→

Unspent → back
margin to the owner, or a refund

DELIVERABLE 3 · SIGNED EPISODE ON 0G STORAGE · SAFE STOP

Every decision is on record. A missing one is never faked.

```
episode bundle → 0G Storage (root 0x45645ef9…7aefef)
  decisions[]  providerRequestId · model · trust · latency
  purchases[]  0G payment tx · agent job score
  manifest_hash · robot_signature   (robot's own key)
```

No answer, late or invalid

The robot holds, retries once, then **SAFE_STOP**. Nothing is invented; this is tested.

Motor fault

It stops locally with no network and no AI; the incident is signed and queued.

WHAT BREAKS IF 0G IS REMOVED

The robot can't decide at a single checkpoint, so it safe-stops: no job done, no agent paid.

Physical AI · Private Computer

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